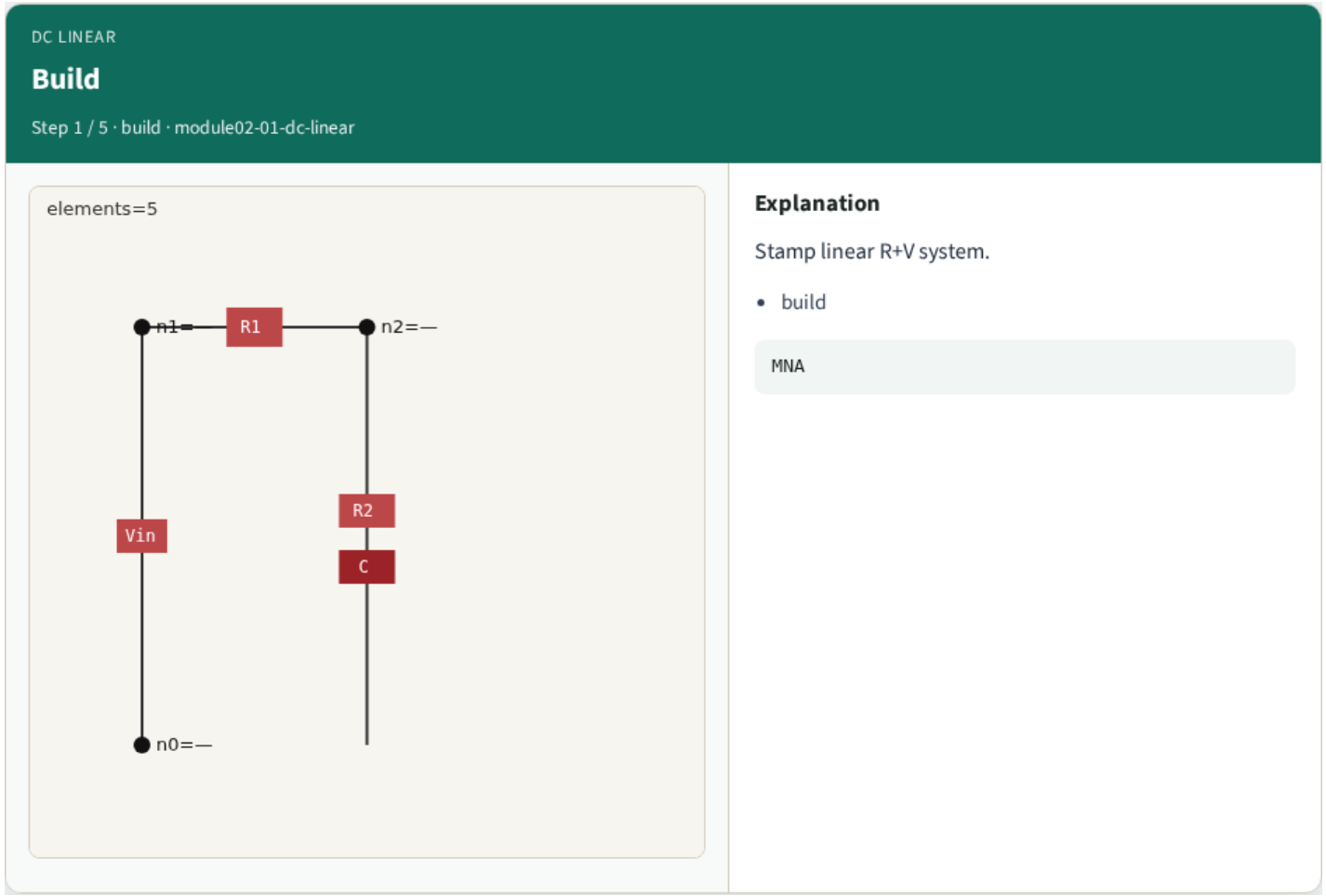
Why DC

Linear DC finds the operating point before Newton or transient analysis

The idea

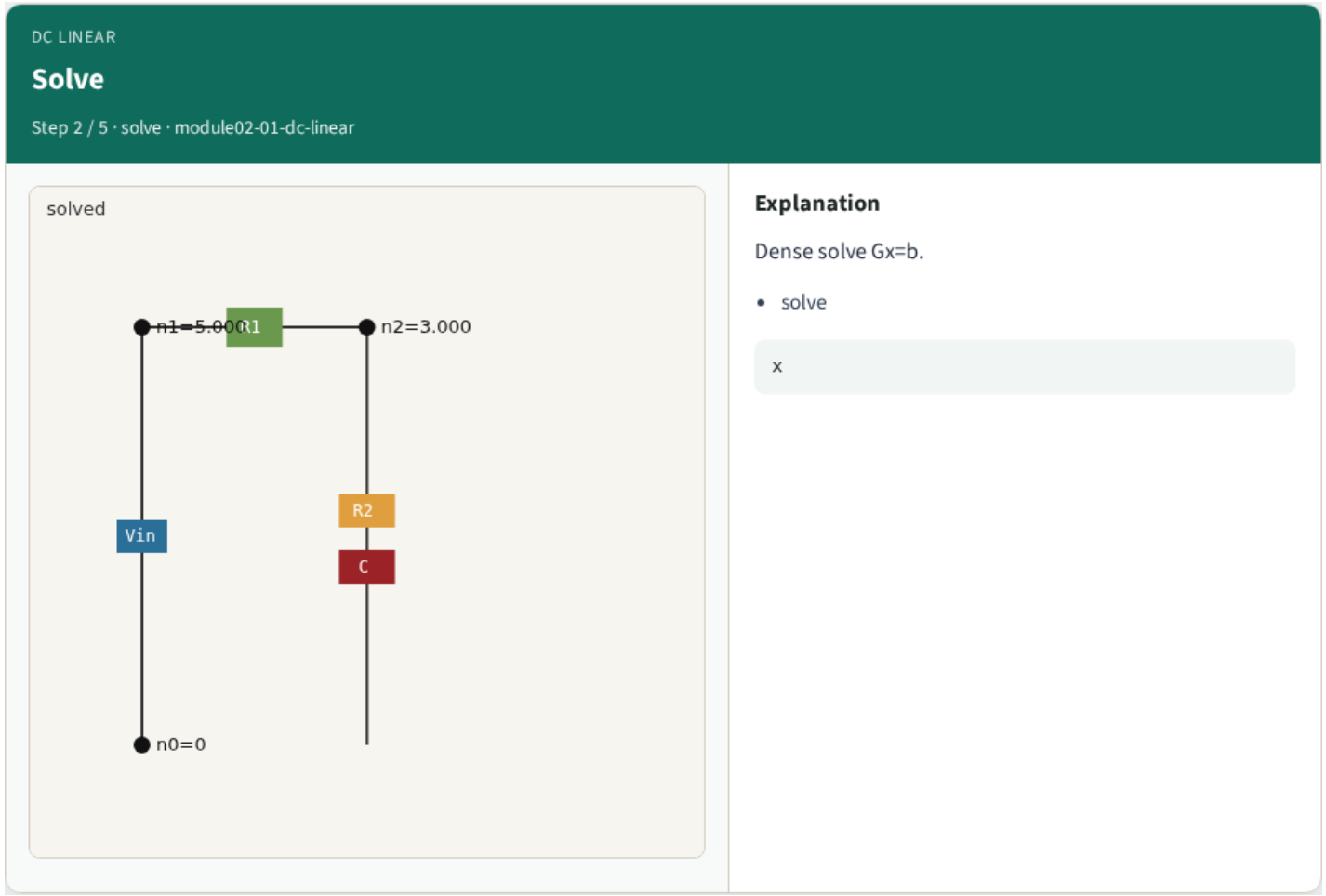
- GOLDEN: V1=5.0, V2=3.0 from the divider 5×3/(2+3)

Build


DC LINEAR
Build
Step 1 / 5 · build · module02-01-dc-linear
elements=5
n1=—
R1
n2=—
R2
Vin
C
n0=—
Explanation
Stamp linear R+V system.
build
MNA

Build

Solve


DC LINEAR
Solve
Step 2 / 5 · solve · module02-01-dc-linear
solved
n1=5.000
R1
n2=3.000
R2
Vin
C
n0=0
Explanation
Dense solve Gx=b.
solve
x

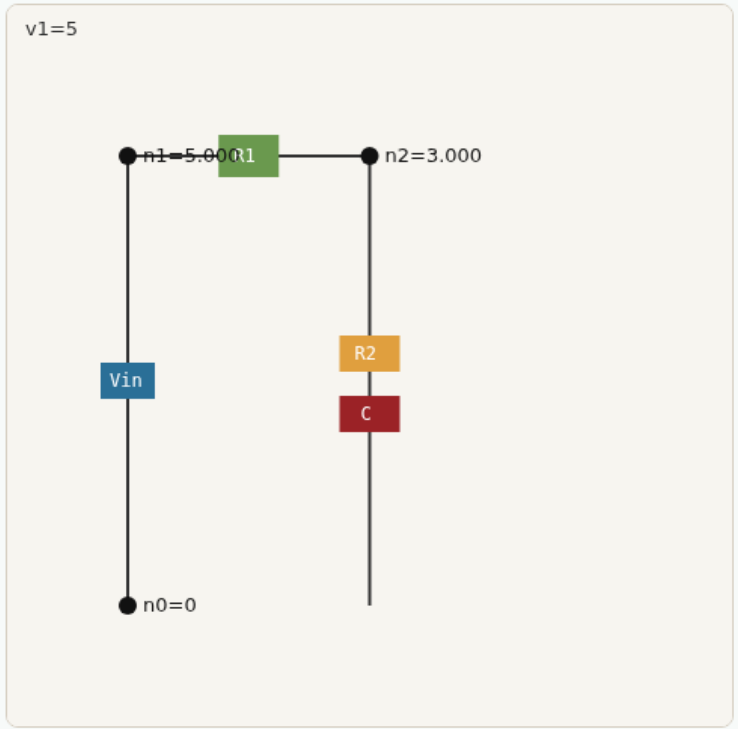
Solve

V1=5

DC LINEAR

V1=5

Step 3 / 5 · v1 · module02-01-dc-linear



Explanation

Node 1 fixed by Vin.

- V1

```
5
```

V1=5

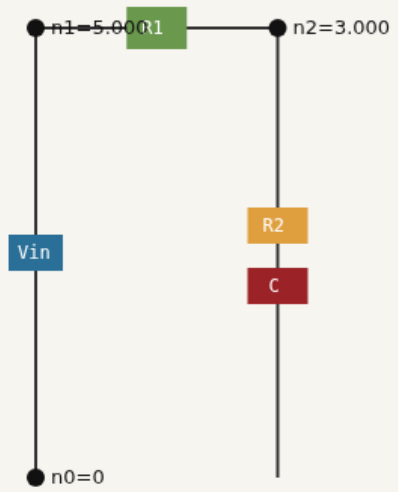
V2=3

DC LINEAR

V2=3

Step 4 / 5 · v2 · module02-01-dc-linear

v2=3.0000000000000004



Explanation

Divider 5·3/(2+3)=3.

- V2

```
3
```

V2=3

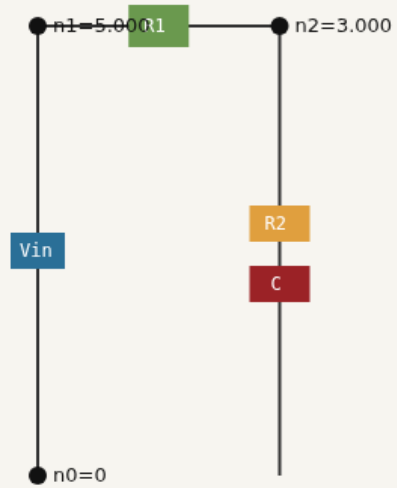
Golden

DC LINEAR

Golden

Step 5 / 5 · g · module02-01-dc-linear

GOLDEN 3



Explanation

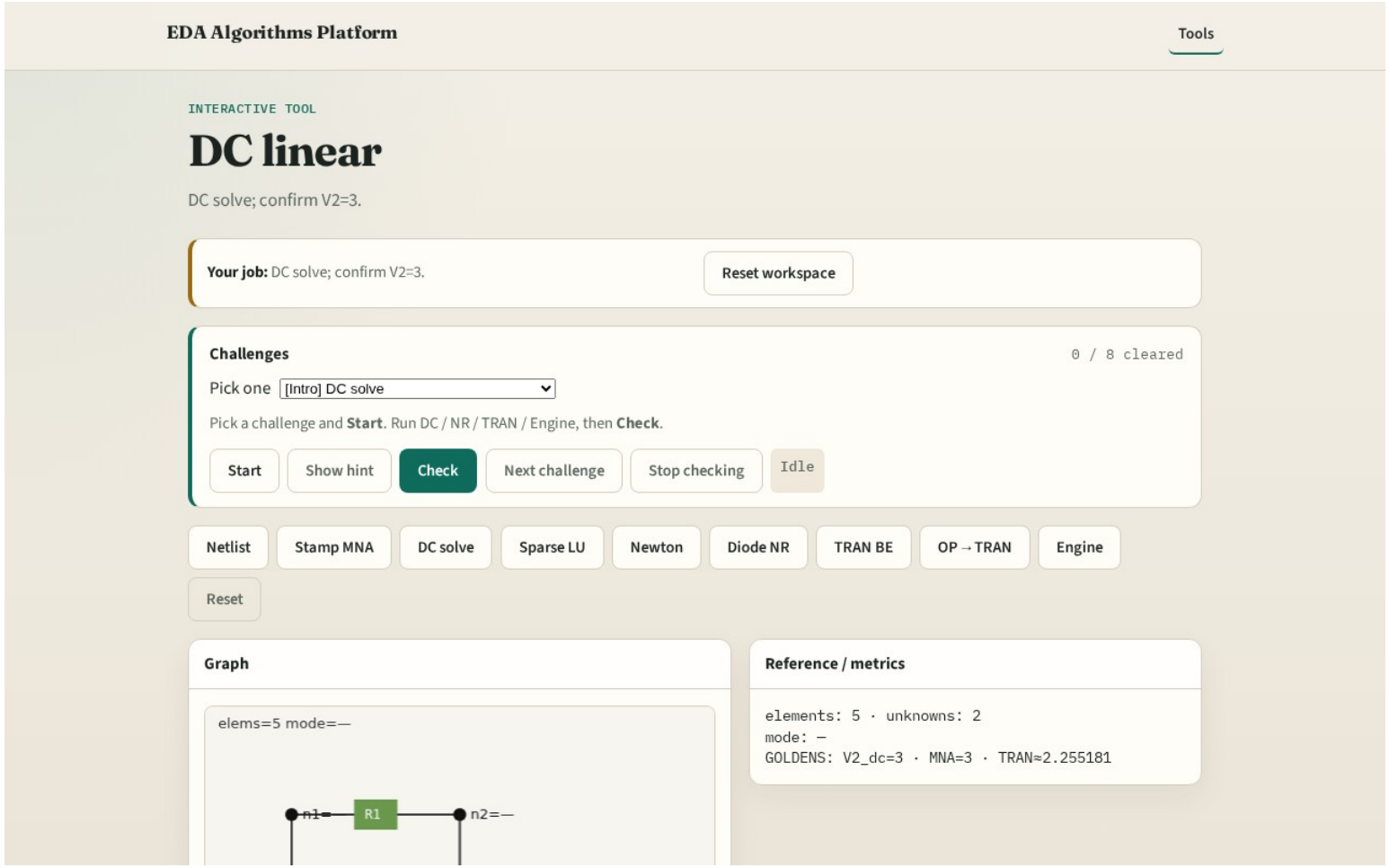
GOLDEN v2_dc=3.

- GOLDEN

```
3
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call dc_linear_solve(); assert abs(volts['2']-3.0)<1e-9 and volts['1']==5.0

Pitfalls

- Including diode in linear DC
- Wrong divider formula

Your turn

- Next: sparse LU literacy on G